Problem 1(b): QED Compton → QCD Gluon Compton

Full Derivation with Diagrams

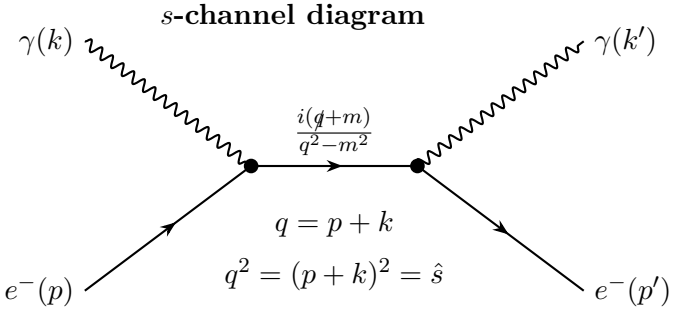
PHYS 7502

1 The Feynman Diagrams: Why s-Channel and u-Channel?

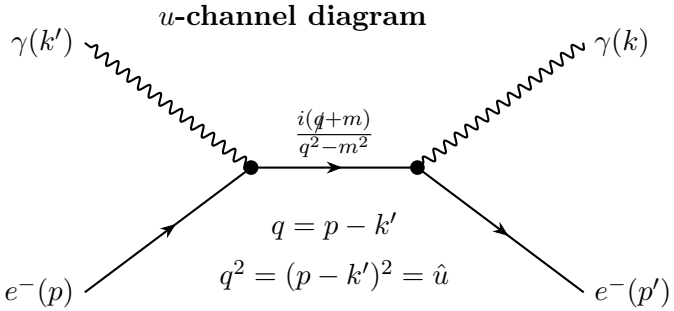
1.1 QED Compton Scattering: $e^-(p) + \gamma(k) \to e^-(p') + \gamma(k')$

There are exactly **two** lowest-order diagrams. Let's draw them and understand why.

s-channel diagram



u-channel diagram



Why These Channels and No Others?

Why s-channel? In the first diagram, the virtual electron carries momentum $q = p + k$ (sum of the two incoming momenta). By definition, $(p + k)^2 = \hat{s}$. The electron absorbs the photon, becoming off-shell, then re-emits.

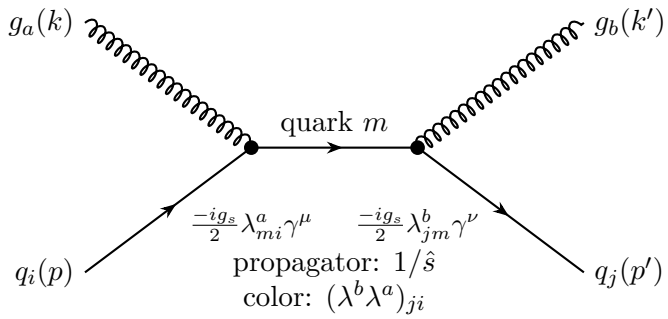
Why u-channel? In the second diagram, the electron emits the outgoing photon *first*, giving the virtual electron momentum $q = p - k'$. By definition, $(p - k')^2 = \hat{u}$.

Why no t-channel? A t-channel diagram would require the momentum transfer $q = p - p'$ flowing through the internal line. This would mean a *photon* propagator connecting the two fermion–photon vertices. But that would require a **three-photon vertex**, which doesn't exist in QED (photons don't carry charge). So there is no t-channel diagram for Compton scattering.

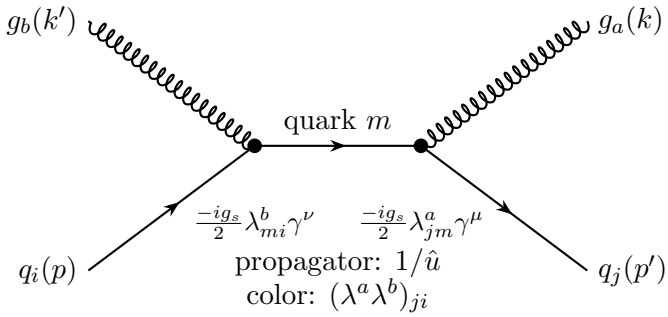
1.2 QCD Gluon Compton: $q(p) + g(k) \to q(p') + g(k')$

For the QCD process, we replace $e^- \to q$ and $\gamma \to g$. The two quark-propagator diagrams have **exactly the same topology**:

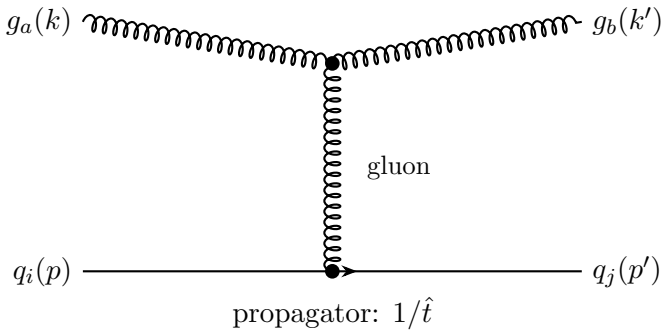
QCD s-channel



QCD u-channel



Third diagram (EXCLUDED from this problem)



Triple-gluon vertex — not in QED!

(Requires f^{abc} structure constants)

Why Does the Third Diagram Exist in QCD but Not QED?

In QED, there is no three-photon vertex because photons are electrically neutral — they don't couple to each other. In QCD, gluons **carry color charge**, so they can interact with each other through the triple-gluon vertex $\propto g_s f^{abc}$. This is a direct consequence of SU(3) being **non-Abelian** ($f^{abc} \neq 0$).

The problem explicitly tells us to ignore this diagram: "you only need to consider the two analogous diagrams involving quark propagators." The full gauge-invariant answer requires all three, but we will discuss that in class.

2 Do We Need to Convert Mandelstam Variables?

The Short Answer

No. The QED and QCD processes have *identical topologies* (spin-$\frac{1}{2}$ + spin-1 $\to$ spin-$\frac{1}{2}$ + spin-1), so the Mandelstam variables map directly:

$$\hat{s} = (p_{\text{fermion}} + p_{\text{boson,in}})^2, \quad \hat{t} = (p_{\text{fermion,in}} - p_{\text{fermion,out}})^2, \quad \hat{u} = (p_{\text{fermion,in}} - p_{\text{boson,out}})^2$$

These definitions are **the same** whether the boson is a photon or a gluon. The propagator channels are $\hat{s}$ and $\hat{u}$ in both cases. The cross section formula $d\hat{\sigma}/d\hat{t} = \overline{|\mathcal{M}|^2}/(16\pi\hat{s}^2)$ is identical.

However, there is a subtlety with **labeling conventions**. Let me be explicit.

2.1 Standard Convention vs. Griffiths Convention

In your QED Compton result, you obtained:

$$\overline{|\mathcal{M}|^2}_{\text{QED}} = -2g_e^4\left(\frac{\hat{s}}{\hat{u}} + \frac{\hat{u}}{\hat{s}}\right) = \frac{-2g_e^4(\hat{s}^2 + \hat{u}^2)}{\hat{s}\hat{u}} \tag{1}$$

where $\hat{s}$ and $\hat{u}$ are the two propagator-channel invariants (as labeled in the diagrams above).

The invariant cross section (differentiating with respect to $\hat{t}$, the momentum transfer):

$$\frac{d\hat{\sigma}}{d\hat{t}} = \frac{\overline{|\mathcal{M}|^2}}{16\pi\hat{s}^2} = \frac{-g_e^4(\hat{s}^2 + \hat{u}^2)}{8\pi\hat{s}^3\hat{u}} \tag{2}$$

Checking Your Notation

In your handwritten notes, you wrote the result with s and t as the two propagator variables. If you used Griffiths' labeling convention (where particles are numbered 1=e^-_{in}, 2=γ_{out}, 3=γ_{in}, 4=e^-_{out}), then what Griffiths calls $t = (p_1 - p_2)^2 = (p_{e,\text{in}} - p_{\gamma,\text{out}})^2$ is the **same** as what the standard convention calls $\hat{u}$.

Bottom line: Your result is correct either way. The physics is in the propagator denominators ($1/\hat{s}$ and $1/\hat{u}$), not in what we call them. For the QCD conversion, just use the **same labeling** as your QED result. No relabeling needed.

For clarity in the rest of this solution, I will use $\hat{s}$ and $\hat{u}$ for the propagator channels (standard convention), with $\hat{t}$ as the momentum transfer. You can substitute your notation at the end.

3 Converting QED → QCD: Step by Step

3.1 Step 1: Write the QCD Amplitudes

Reading backward along the fermion line and applying the QCD Feynman rules (Griffiths Sec. 9.1):

s-channel (a quark absorbs gluon a, propagates, emits gluon b):

$$i\mathcal{M}_s = \bar{u}(p')\left(\frac{-ig_s}{2}\lambda^b_{jm}\gamma^\nu\right)\varepsilon'^*_\nu\,\frac{i(\not{p}+\not{k})}{(p+k)^2}\,\varepsilon_\mu\left(\frac{-ig_s}{2}\lambda^a_{mi}\gamma^\mu\right)u(p) \tag{3}$$

The color matrices chain as: first vertex gives λ^a_{mi} (color $i \to m$), second gives λ^b_{jm} (color $m \to j$). Summing over the intermediate color m:

$$\sum_m \lambda^b_{jm}\lambda^a_{mi} = (\lambda^b\lambda^a)_{ji} \tag{4}$$

So the amplitude factorizes into color × Dirac:

$$\mathcal{M}_s = \frac{g_s^2}{4}(\lambda^b\lambda^a)_{ji} \;\times\; \underbrace{\left[\frac{-1}{\hat{s}}\bar{u}(p')\not{\varepsilon}'^*(\not{p}+\not{k})\not{\varepsilon}\,u(p)\right]}_{D_s\text{ (same Dirac structure as QED with } e^2=1)} \tag{5}$$

u-channel (b quark emits gluon b first, propagates, absorbs gluon a):

By the same logic, the color matrices chain as $\lambda^a_{jm}\lambda^b_{mi} = (\lambda^a\lambda^b)_{ji}$:

$$\mathcal{M}_u = \frac{g_s^2}{4}(\lambda^a\lambda^b)_{ji} \;\times\; \underbrace{\left[\frac{-1}{\hat{u}}\bar{u}(p')\not{\varepsilon}(\not{p}-\not{k}')\not{\varepsilon}'^*\,u(p)\right]}_{D_u\text{ (same Dirac structure as QED with } e^2=1)} \tag{6}$$

Key Observation: Factorization

The color part (λ matrices) and the spin part (γ matrices, spinors) act on **different vector spaces** (3-dimensional color space vs. 4-dimensional spinor space). They factorize completely:

$$\mathcal{M} = \underbrace{(\text{color factor})}_{3\times3\text{ matrices}} \times \underbrace{(\text{coupling})}_{\text{numbers}} \times \underbrace{(\text{Dirac structure})}_{4\times4\text{ matrices}}$$

The Dirac structure D_s and D_u are **exactly the same** as in QED Compton (with $e^2 \to 1$). So all the trace algebra that gave $-2(\hat{s}/\hat{u} + \hat{u}/\hat{s})$ carries over unchanged.

3.2 Step 2: Square the Amplitude

The total amplitude is $\mathcal{M} = \mathcal{M}_s + \mathcal{M}_u$:

$$|\mathcal{M}|^2 = |\mathcal{M}_s|^2 + |\mathcal{M}_u|^2 + 2\,\mathrm{Re}(\mathcal{M}_s^*\mathcal{M}_u) \tag{7}$$

Using the factorized forms (5)–(6):

$$|\mathcal{M}_s|^2 = \frac{g_s^4}{16} \, |(\lambda^b \lambda^a)_{ji}|^2 \;\times\; |D_s|^2 \tag{8}$$

$$|\mathcal{M}_u|^2 = \frac{g_s^4}{16} \, |(\lambda^a \lambda^b)_{ji}|^2 \;\times\; |D_u|^2 \tag{9}$$

$$\mathcal{M}_s^* \mathcal{M}_u = \frac{g_s^4}{16} \, (\lambda^b \lambda^a)^*_{ji} (\lambda^a \lambda^b)_{ji} \;\times\; D_s^* D_u \tag{10}$$

3.3 Step 3: Spin/Polarization Sums

Now sum over final spins and polarizations, average over initial ones ($\frac{1}{4}$: 2 quark spins $\times$ 2 gluon polarizations). The Dirac parts give (from the QED calculation with $e^2 = 1$):

$$\frac{1}{4} \sum_{\text{spins,pols}} |D_s|^2 = -\frac{2\hat{u}}{\hat{s}} \tag{11}$$

$$\frac{1}{4} \sum_{\text{spins,pols}} |D_u|^2 = -\frac{2\hat{s}}{\hat{u}} \tag{12}$$

$$\frac{1}{4} \sum_{\text{spins,pols}} D_s^* D_u = 0 \qquad \text{(no spin interference for } m = 0\text{)} \tag{13}$$

Why the Spin Interference Vanishes

In the massless limit, Compton scattering conserves helicity. The s-channel and u-channel diagrams contribute to *different helicity configurations* of the fermion–boson system. Different helicity amplitudes are orthogonal, so their interference is zero. This is exactly what happens in QED (Lancaster & Blundell Eq. 40.16), and since the Dirac structure is identical in QCD, the same cancellation occurs.

Important: This means the color interference term (10) drops out — even though $(\lambda^b \lambda^a)^*_{ji} (\lambda^a \lambda^b)_{ji} \neq 0$ in general, it multiplies the zero Dirac trace $\sum D_s^* D_u = 0$.

3.4 Step 4: Color Sums and Averages

The spin-averaged result (before color) is:

$$\frac{1}{4} \sum_{\text{spins,pols}} |\mathcal{M}|^2 = \frac{g_s^4}{16} \left[|(\lambda^b \lambda^a)_{ji}|^2 \cdot \left(-\frac{2\hat{u}}{\hat{s}} \right) + |(\lambda^a \lambda^b)_{ji}|^2 \cdot \left(-\frac{2\hat{s}}{\hat{u}} \right) \right] \tag{14}$$

Now we need to average over initial colors and sum over final colors:

$$\text{Average over initial quark color } i: \quad \frac{1}{N_c} = \frac{1}{3} \tag{15}$$

$$\text{Average over initial gluon color } a: \quad \frac{1}{N_c^2 - 1} = \frac{1}{8} \tag{16}$$

$$\text{Sum over final quark color } j \text{ and gluon color } b \tag{17}$$

Define the color-averaged factor for $|s\text{-channel}|^2$:

$$\mathcal{C}_s \;\equiv\; \frac{1}{3 \cdot 8} \sum_{i,j=1}^{3} \sum_{a,b=1}^{8} |(\lambda^b \lambda^a)_{ji}|^2 \tag{18}$$

3.4.1 Evaluating $\mathcal{C}_s$ Step by Step

Step 4a: Write out $|(\lambda^b\lambda^a)_{ji}|^2$ using Hermiticity of the λ matrices:

$$\begin{aligned}|(\lambda^b\lambda^a)_{ji}|^2 &= (\lambda^b\lambda^a)_{ji}\,[(\lambda^b\lambda^a)_{ji}]^* \\ &= (\lambda^b\lambda^a)_{ji}\,(\lambda^b\lambda^a)^*_{ji} \\ &= (\lambda^b\lambda^a)_{ji}\,(\lambda^{a\dagger}\lambda^{b\dagger})_{ij} \\ &= (\lambda^b\lambda^a)_{ji}\,(\lambda^a\lambda^b)_{ij} \qquad (\text{since } \lambda^{a\dagger}=\lambda^a)\end{aligned} \tag{19}$$

Step 4b: Sum over i and j — this produces a trace:

$$\sum_{i,j}(\lambda^b\lambda^a)_{ji}(\lambda^a\lambda^b)_{ij} = \sum_j\Big[(\lambda^b\lambda^a)(\lambda^a\lambda^b)\Big]_{jj} = \mathrm{Tr}\Big[\lambda^b\lambda^a\lambda^a\lambda^b\Big] \tag{20}$$

So:

$$\mathcal{C}_s = \frac{1}{24}\sum_{a=1}^{8}\sum_{b=1}^{8}\mathrm{Tr}\Big[\lambda^b\lambda^a\lambda^a\lambda^b\Big] \tag{21}$$

Step 4c: Evaluate the inner sum $\sum_a \lambda^a\lambda^a$ using the **quadratic Casimir**:

$$\sum_{a=1}^{8}T^aT^a = C_F\,\mathbb{I}_3 \qquad \text{where} \quad T^a=\frac{\lambda^a}{2}, \quad C_F=\frac{N_c^2-1}{2N_c}=\frac{4}{3} \tag{22}$$

Therefore:

$$\boxed{\sum_{a=1}^{8}\lambda^a\lambda^a = 4C_F\,\mathbb{I}_3 = \frac{16}{3}\,\mathbb{I}_3} \tag{23}$$

Quick Verification of the Casimir

Check the $(1,1)$ element: $\sum_a(\lambda^a\lambda^a)_{11}=\sum_a\sum_k|\lambda^a_{1k}|^2$. Going through each λ^a:
λ^1: $|\lambda^1_{12}|^2=1$. λ^2: $|\lambda^2_{12}|^2=1$. λ^3: $|\lambda^3_{11}|^2=1$. λ^4: $|\lambda^4_{13}|^2=1$. λ^5: $|\lambda^5_{13}|^2=1$. λ^6: 0.
λ^7: 0. λ^8: $|1/\sqrt{3}|^2=1/3$.
Total: $1+1+1+1+1+0+0+\frac{1}{3}=\frac{16}{3}$. ✓

Step 4d: Substitute (23) into (21):

$$\mathcal{C}_s = \frac{1}{24}\sum_{b=1}^{8}\mathrm{Tr}\left[\lambda^b\cdot\frac{16}{3}\mathbb{I}\cdot\lambda^b\right] = \frac{1}{24}\cdot\frac{16}{3}\sum_{b=1}^{8}\mathrm{Tr}(\lambda^b\lambda^b) \tag{24}$$

Step 4e: Use the trace identity $\mathrm{Tr}(\lambda^a\lambda^b)=2\delta^{ab}$:

$$\sum_{b=1}^{8}\mathrm{Tr}(\lambda^b\lambda^b) = \sum_{b=1}^{8}2\delta^{bb} = 2\times 8 = 16 \tag{25}$$

Step 4f: Final result:

$$\mathcal{C}_s = \frac{1}{24}\cdot\frac{16}{3}\cdot 16 = \frac{256}{72} = \frac{32}{9} \tag{26}$$

By the identical argument with $a\leftrightarrow b$, $\mathcal{C}_u=\mathcal{C}_s=32/9$.

Color Factor for $qg \to qg$ (Both Channels)

$$\boxed{\mathcal{C}_s = \mathcal{C}_u = \frac{32}{9}} \tag{27}$$

Anatomy: $\dfrac{32}{9} = \underbrace{\dfrac{1}{3}}_{\text{avg } i} \times \underbrace{\dfrac{1}{8}}_{\text{avg } a} \times \underbrace{\dfrac{256}{3}}_{\sum_{i,j,a,b} |(\lambda^b \lambda^a)_{ji}|^2}$

3.5 Step 5: Combine Everything

Putting together coupling ($g_s^4/16$), Dirac structure, and color factor:

$$\begin{aligned}
\overline{|\mathcal{M}|^2}_{\text{QCD}} &= \frac{g_s^4}{16}\left[\mathcal{C}_s \cdot \left(-\frac{2\hat{u}}{\hat{s}}\right) + \mathcal{C}_u \cdot \left(-\frac{2\hat{s}}{\hat{u}}\right)\right] \\
&= \frac{g_s^4}{16} \cdot \frac{32}{9} \cdot (-2)\left(\frac{\hat{u}}{\hat{s}} + \frac{\hat{s}}{\hat{u}}\right) \\
&= \frac{-4g_s^4}{9}\left(\frac{\hat{u}}{\hat{s}} + \frac{\hat{s}}{\hat{u}}\right)
\end{aligned} \tag{28}$$

Cross-check the structure: Compare with QED: $\overline{|\mathcal{M}|^2}_{\text{QED}} = -2g_e^4(\hat{u}/\hat{s} + \hat{s}/\hat{u})$. The ratio is:

$$\frac{\overline{|\mathcal{M}|^2}_{\text{QCD}}}{\overline{|\mathcal{M}|^2}_{\text{QED}}} = \frac{-4g_s^4/9}{-2g_e^4} = \frac{2g_s^4}{9g_e^4} = \frac{2\alpha_s^2}{9\alpha^2} \tag{29}$$

This makes sense: coupling $\alpha^2 \to \alpha_s^2$, times a color factor $2/9$.

3.6 Step 6: The Cross Section

Using $g_s^2 = 4\pi\alpha_s$, so $g_s^4 = 16\pi^2\alpha_s^2$:

$$\overline{|\mathcal{M}|^2}_{\text{QCD}} = \frac{-4 \cdot 16\pi^2\alpha_s^2}{9}\left(\frac{\hat{u}}{\hat{s}} + \frac{\hat{s}}{\hat{u}}\right) = \frac{-64\pi^2\alpha_s^2}{9} \cdot \frac{(\hat{s}^2 + \hat{u}^2)}{\hat{s}\hat{u}} \tag{30}$$

The invariant cross section:

$$\frac{d\hat{\sigma}}{d\hat{t}} = \frac{\overline{|\mathcal{M}|^2}}{16\pi\hat{s}^2} = \frac{-64\pi^2\alpha_s^2(\hat{s}^2 + \hat{u}^2)}{9 \cdot 16\pi \cdot \hat{s}^3\hat{u}} = \frac{-4\pi\alpha_s^2(\hat{s}^2 + \hat{u}^2)}{9\,\hat{s}^3\hat{u}} \tag{31}$$

Final Answer: $qg \to qg$ Cross Section (Two Diagrams)

$$\boxed{\left.\frac{d\hat{\sigma}}{d\hat{t}}\right|_{\text{2 diag}} = \frac{-4\pi\alpha_s^2(\hat{s}^2+\hat{u}^2)}{9\,\hat{s}^3\hat{u}}} \tag{32}$$

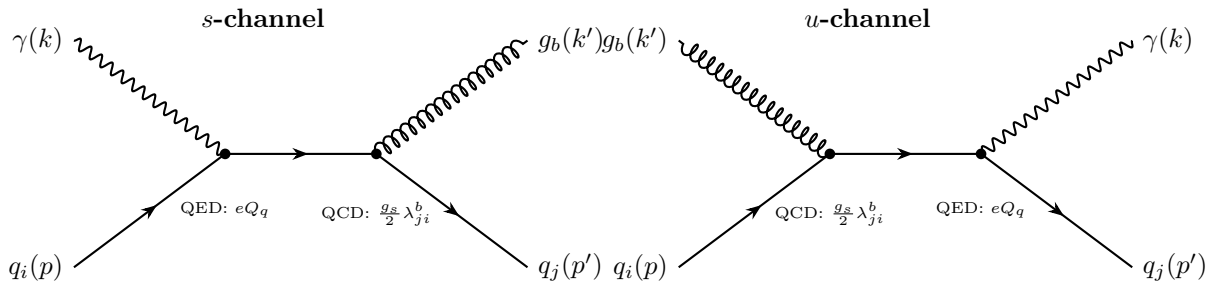
In your notation (if your propagator variables are $\hat{s}$ and $\hat{t}$):

$$\boxed{\left.\frac{d\hat{\sigma}}{d\hat{u}}\right|_{\text{2 diag}} = \frac{-4\pi\alpha_s^2(\hat{s}^2+\hat{t}^2)}{9\,\hat{s}^3\hat{t}}} \tag{33}$$

Or equivalently, simply take your QED result and multiply:

$$\left(\frac{d\hat{\sigma}}{d\hat{t}}\right)_{\text{QCD}} = \frac{2\alpha_s^2}{9\alpha^2} \times \left(\frac{d\hat{\sigma}}{d\hat{t}}\right)_{\text{QED}} \tag{34}$$

4 Bonus: The Easier Alternative $q\gamma \to qg$



Both diagrams have color factor $\frac{1}{2}\lambda^b_{ji}$ (the photon is color-blind). Coupling: $g_e^4 \to (eQ_q)^2(g_s/2)^2 = e^2Q_q^2g_s^2/4$.

Color average (only average over quark color; photon has no color):

$$\mathcal{C} = \frac{1}{3}\sum_{i,j,b}|\lambda^b_{ji}|^2 = \frac{1}{3}\sum_b \mathrm{Tr}(\lambda^b\lambda^b) = \frac{1}{3}\times 16 = \frac{16}{3}$$

$q\gamma \to qg$ Cross Section

$$\boxed{\frac{d\hat{\sigma}}{d\hat{t}}(q\gamma \to qg) = \frac{-8\pi\alpha\alpha_s Q_q^2(\hat{s}^2+\hat{u}^2)}{3\,\hat{s}^3\hat{u}}} \tag{35}$$

5 Summary

Complete Conversion Table

Process	Coupling in $\lvert\mathcal{M}\rvert^2$	Color Factor	Overall Multiplier
$e\gamma \to e\gamma$	e^4	1	e^4
$q\gamma \to qg$	$e^2Q_q^2g_s^2/4$	16/3	$4e^2Q_q^2g_s^2/3$
$qg \to qg$ (2 diag)	$g_s^4/16$	32/9	$2g_s^4/9$